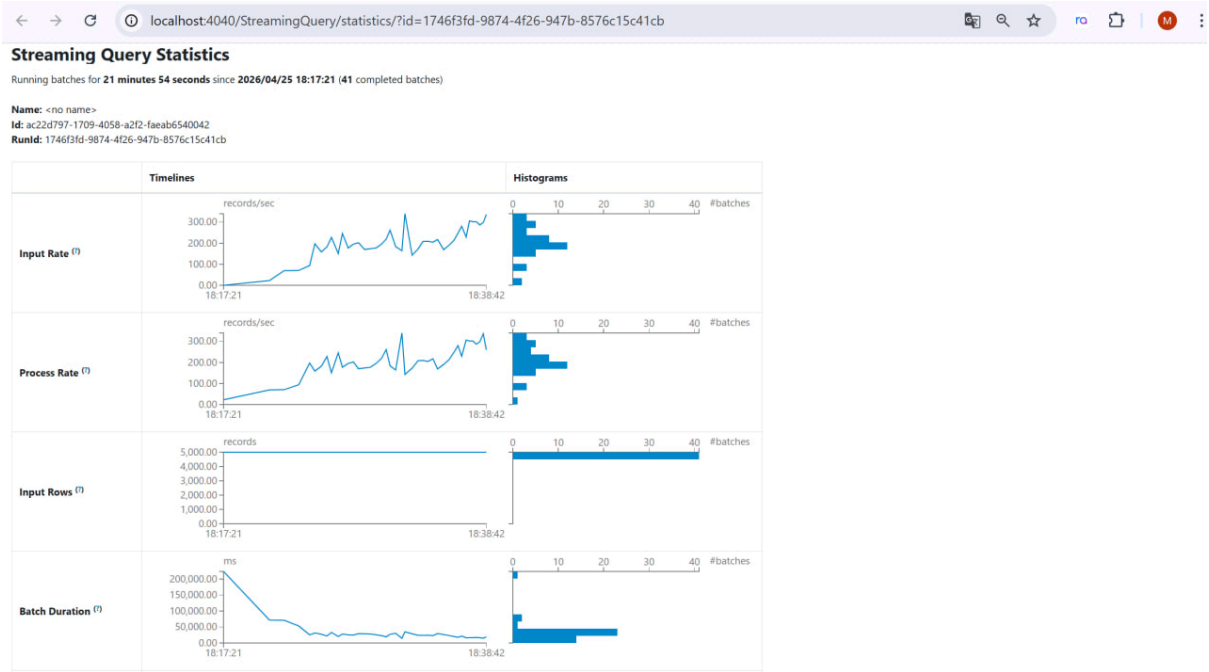
1. Дані успішно завантажені в Kafka

```
        found mysql#mysql-connector-java;8.0.28 in central
        found com.google.protobuf#protobuf-java;3.11.4 in central
:: resolution report :: resolve 321ms :: artifacts dl 7ms
        :: modules in use:
        com.google.code.findbugs#jsr305;3.0.0 from central in [default]
        com.google.protobuf#protobuf-java;3.11.4 from central in [default]
        commons-logging#commons-logging;1.1.3 from central in [default]
        mysql#mysql-connector-java;8.0.28 from central in [default]
        org.apache.commons#commons-pool2;2.11.1 from central in [default]
        org.apache.hadoop#hadoop-client-api;3.3.4 from central in [default]
        org.apache.hadoop#hadoop-client-runtime;3.3.4 from central in [default]
        org.apache.kafka#kafka-clients;3.4.1 from central in [default]
        org.apache.spark#spark-sql-kafka-0-10_2.12;3.5.0 from central in [default]
        org.apache.spark#spark-token-provider-kafka-0-10_2.12;3.5.0 from central in [default]
        org.lz4#lz4-java;1.8.0 from central in [default]
        org.slf4j#slf4j-api;2.0.7 from central in [default]
        org.xerial.snappy#snappy-java;1.1.10.3 from central in [default]
        ---------------------------------------------------------------------
        |                  |            modules            ||   artifacts   |
        |       conf       | number| search|dwnlded|evicted|| number|dwnlded|
        ---------------------------------------------------------------------
        |      default     |   13  |   0   |   0   |   0   ||   13  |   0   |
        ---------------------------------------------------------------------
:: retrieving :: org.apache.spark#spark-submit-parent-529c799c-585f-4a49-8dfd-8b6635ebd6fa
        confs: [default]
        0 artifacts copied, 13 already retrieved (0kB/6ms)
26/04/25 18:16:10 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where
 applicable
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Запитуємо дані у MySQL (це буде швидко)...
Відправляємо в Кафку...
Успішно! Кафка наповнена.
(airflow_env) marichka@DESKTOP-SFONK0I:~/spark_streaming_project$
```

2. Spark в реальному часі обробляє дані, робить Join та рахує середнє.

```
marichka@DESKTOP-SFONK0I

26/04/25 18:34:08 WARN HDFSBackedStateStoreProvider: The state for version 26 doesn't exist in loadedMaps. Reading snaps
hot file and delta files if needed...Note that this is normal for the first batch of starting query.
26/04/25 18:34:08 WARN HDFSBackedStateStoreProvider: The state for version 26 doesn't exist in loadedMaps. Reading snaps
hot file and delta files if needed...Note that this is normal for the first batch of starting query.
--- Processing Batch 27 ---
+------------------+------+----+-----------+------------------+-----------------+-----------------------+
|sport             |medal |sex |country_noc|avg_height        |avg_weight       |timestamp              |
+------------------+------+----+-----------+------------------+-----------------+-----------------------+
|Sailing           |nan   |NULL|SRI        |173.25            |74.75            |2026-04-25 18:34:21.841|
|Rowing            |Silver|NULL|GER        |184.58333333333334|82.08333333333333|2026-04-25 18:34:21.841|
|EquestrianEventing|nan   |NULL|GER        |177.30769230769232|69.6923076923077 |2026-04-25 18:34:21.841|
|Weightlifting     |nan   |NULL|SYR        |177.75            |102.25           |2026-04-25 18:34:21.841|
|Luge              |Silver|NULL|GER        |175.5             |72.0             |2026-04-25 18:34:21.841|
|Diving            |nan   |NULL|GER        |166.125           |62.125           |2026-04-25 18:34:21.841|
|EquestrianJumping |nan   |NULL|SYR        |168.0             |72.0             |2026-04-25 18:34:21.841|
|Badminton         |nan   |NULL|SRI        |169.66666666666666|66.0             |2026-04-25 18:34:21.841|
|CanoeMarathon     |Silver|NULL|GER        |180.0             |89.0             |2026-04-25 18:34:21.841|
|Fencing           |Gold  |NULL|GER        |166.0             |63.0             |2026-04-25 18:34:21.841|
|Shooting          |nan   |NULL|SYR        |172.0             |84.66666666666667|2026-04-25 18:34:21.841|
|FigureSkating     |nan   |NULL|GER        |167.6153846153846 |58.07692307692308|2026-04-25 18:34:21.841|
|Luge              |Gold  |NULL|GER        |167.0             |63.0             |2026-04-25 18:34:21.841|
|EquestrianJumping |Bronze|NULL|GER        |166.0             |64.0             |2026-04-25 18:34:21.841|
|EquestrianJumping |nan   |NULL|GER        |170.57142857142858|68.28571428571429|2026-04-25 18:34:21.841|
|Athletics         |Gold  |NULL|GER        |178.6             |77.4             |2026-04-25 18:34:21.841|
|Athletics         |nan   |NULL|SRI        |171.25757575757575|59.09090909090909|2026-04-25 18:34:21.841|
|AlpineSkiing      |nan   |NULL|GER        |172.02777777777777|64.33333333333333|2026-04-25 18:34:21.841|
|CyclingRoad       |nan   |NULL|GER        |175.35714285714286|70.57142857142857|2026-04-25 18:34:21.841|
|Sailing           |Silver|NULL|GER        |173.66666666666666|78.0             |2026-04-25 18:34:21.841|
+------------------+------+----+-----------+------------------+-----------------+-----------------------+
only showing top 20 rows
```



3. Фінальні дані успішно записані в базу для подальшого аналізу.

```
mysql> SELECT * FROM marichka_streaming_results LIMIT 10;
+--------------------+--------+------+-------------+-------------------+-------------------+---------------------+
| sport              | medal  | sex  | country_noc | avg_height        | avg_weight        | timestamp           |
+--------------------+--------+------+-------------+-------------------+-------------------+---------------------+
| Athletics          | Gold   | NULL | BRN         |               166 |                56 | 2026-04-25 17:33:42 |
| Wrestling          | nan    | NULL | AZE         | 172.7872340425532 | 76.59574468085107 | 2026-04-25 17:33:42 |
| RhythmicGymnastics | nan    | NULL | CPV         |            167.75 |             57.75 | 2026-04-25 17:33:42 |
| Basketball         | nan    | NULL | ANG         |           191.875 | 90.36111111111111 | 2026-04-25 17:33:42 |
| Taekwondo          | nan    | NULL | BIH         |               185 |                94 | 2026-04-25 17:33:42 |
| Wrestling          | nan    | NULL | ALG         | 174.55555555555554 | 72.66666666666667 | 2026-04-25 17:33:42 |
| Triathlon          | nan    | NULL | AZE         |               187 |                74 | 2026-04-25 17:33:42 |
| AlpineSkiing       | nan    | NULL | ALG         |               169 |                59 | 2026-04-25 17:33:42 |
| Athletics          | nan    | NULL | ANT         | 176.0096153846154 | 66.59615384615384 | 2026-04-25 17:33:42 |
| Wrestling          | Silver | NULL | ARM         |            168.75 |             73.25 | 2026-04-25 17:33:42 |
+--------------------+--------+------+-------------+-------------------+-------------------+---------------------+
10 rows in set (0.07 sec)
```